Explainable Hybrid Music Recommendation Using User Behavior, Audio Features, and Semantic Tag Modeling

Master's Thesis Proposal

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Abstract

Music recommendation systems are central to modern streaming platforms because they support personalized listening experiences, user retention, and music discovery. Existing recommendation systems commonly rely on collaborative filtering or content-based recommendation independently, limiting their ability to capture the diverse factors influencing user music preferences. Furthermore, many recommendation systems operate as black-box models and provide limited transparency regarding why specific tracks are recommended.

This thesis proposes an explainable hybrid music recommendation framework integrating user interaction behavior, audio/acoustic features, and semantic tag metadata. The proposed framework combines collaborative filtering, acoustic feature modeling, semantic metadata encoding, multimodal fusion, and explainable ranking generation. The system aims to improve recommendation accuracy, personalization quality, interpretability, and robustness under sparse-data and cold-start conditions.

The proposed study evaluates multiple recommendation strategies, including collaborative filtering, content-based audio recommendation, semantic tag recommendation, simple hybrid fusion, and an explainable attention-based multimodal hybrid model. Experiments are conducted using integrated music recommendation datasets containing user listening history, Spotify-style audio features, and semantic metadata from tags, genres, moods, and artist descriptors.

Evaluation is performed using Top-K recommendation metrics including Precision@K, Recall@K, NDCG@K, MAP@K, and Hit Rate@K. Additional experiments analyze audio-feature contribution, semantic metadata utility, fusion-strategy optimization, explanation fidelity, and robustness under cold-start and sparse-data conditions. The expected results demonstrate that multimodal explainable hybrid recommendation significantly improves recommendation reliability and transparency compared with single-source recommendation systems.

Introduction and Background

Modern music streaming platforms generate extremely large music catalogs containing millions of tracks and users. Recommendation systems therefore play a critical role in helping users discover relevant music efficiently. Collaborative filtering approaches have traditionally dominated music recommendation because they effectively model user–item interaction behavior. However, collaborative approaches struggle when interaction data are sparse or when new users and new tracks are introduced.

Content-based recommendation approaches address some of these limitations by utilizing acoustic and audio-related information such as tempo, energy, danceability, loudness,

valence, and acousticness. Although these methods capture musical similarity, they frequently overlook broader behavioral patterns and contextual listening preferences.

Semantic metadata such as genres, moods, artist styles, and user-generated tags provide an additional layer of information capable of improving recommendation relevance and interpretability. Despite this, many current recommendation systems fail to effectively integrate behavioral, acoustic, and semantic information simultaneously.

Another major challenge is the lack of explainability in recommendation systems. Users often receive recommendations without understanding why particular songs are suggested. Explainability is increasingly important in intelligent recommendation systems because transparent recommendations improve user trust, satisfaction, and perceived system reliability.

This research proposes an explainable hybrid recommendation framework that integrates user listening behavior, audio features, and semantic metadata into a unified multimodal recommendation architecture suitable for intelligent decision-support applications.

Problem Statement

Current music recommendation systems face several major limitations:

- Collaborative filtering systems suffer under sparse interaction data and cold-start conditions.
- Content-based systems capture acoustic similarity but frequently fail to model broader listening behavior.
- Semantic recommendation methods improve interpretability but often lack sufficient personalization.
- Existing recommendation systems commonly operate as black boxes with minimal explanation capability.
- Many systems do not provide robustness against missing metadata or incomplete user interaction histories.

The central research problem is therefore:

How can an explainable hybrid music recommendation system effectively integrate user behavior, audio features, and semantic metadata to improve recommendation accuracy, interpretability, and robustness under real-world streaming conditions?

Research Objectives

This research will pursue the following specific objectives:

1. Develop an explainable hybrid music recommendation framework integrating collaborative, acoustic, and semantic information.
2. Compare multimodal recommendation performance against collaborative-only, audio-only, semantic-only, and simple hybrid baselines.
3. Analyze the contribution of user behavior in capturing personalized listening preferences.
4. Evaluate the role of audio/acoustic features in content-aware recommendation.
5. Investigate the effect of semantic metadata on recommendation relevance and interpretability.
6. Compare different multimodal fusion strategies.
7. Develop an explanation-generation layer for transparent recommendation ranking.
8. Evaluate robustness under sparse-user and cold-start scenarios.
9. Produce a publication-ready evaluation framework suitable for Expert Systems with Applications.

Research Questions

This thesis will be guided by the following research questions:

1. **RQ1**: How effectively can a hybrid music recommendation framework integrating user behavior, audio features, and semantic tags improve recommendation accuracy compared with single-source recommendation models?
2. **RQ2**: To what extent can user–item interaction patterns capture personalized listening preferences under varying user activity levels?
3. **RQ3**: How do acoustic attributes contribute to improving content-aware music recommendation performance?
4. **RQ4**: How effectively can genre, mood, artist, and tag-based semantic metadata enhance recommendation relevance and interpretability?
5. **RQ5**: Which multimodal fusion strategy provides the best balance between recommendation accuracy and interpretability?
6. **RQ6:** How can recommendation explanations generated from behavioral, acoustic, and semantic evidence improve transparency and user trust?
7. **RQ7:** How robust is the proposed explainable hybrid recommendation framework under sparse interaction data, cold-start users, cold-start tracks, and incomplete metadata conditions?

Theoretical Framework

The theoretical underpinning of this research integrates:

- Collaborative Filtering Theory for user–item preference modeling.
- Content-Based Recommendation Theory using audio similarity.
- Semantic Recommendation and Tag-Based Retrieval.
- Explainable Artificial Intelligence (XAI) for recommendation transparency.
- Multimodal Representation Learning and Fusion.
- Human-Centered Recommendation Systems.
- Top-K Ranking Optimization.
- Cold-Start and Sparse-Data Recommendation Theory.

Methodology

6.1 Research Design

- collaborative filtering recommendation,
- audio feature recommendation,
- semantic recommendation,
- multimodal hybrid fusion,
- explainable ranking generation,
- comparative evaluation against baseline systems.

The proposed architecture contains:

1. User–item interaction encoder
2. Audio feature encoder
3. Semantic metadata encoder
4. Fusion-based ranking layer
5. Explainability layer

6.2 Dataset Sources

Primary datasets include:

- Million Song Dataset
- Spotify Audio Features Dataset
- Last.fm User Listening Dataset
- Integrated Kaggle Hybrid Music Recommendation Dataset

The datasets provide:

- user listening history,
- track metadata,
- Spotify-style acoustic features,
- genre and mood labels,
- user-generated tags.

6.3 Data Preprocessing

The preprocessing pipeline includes:

- duplicate removal,
- user–track alignment,
- interaction normalization,
- sparse matrix construction,
- audio feature normalization,
- TF-IDF semantic encoding,
- train/validation/test splitting.

6.4 Feature Engineering

Behavioral Features

- play count,
- interaction frequency,
- user embeddings,

- implicit-feedback scores.

Audio Features

- tempo,
- energy,
- loudness,
- danceability,
- valence,
- acousticness,
- liveness,
- instrumentalness.

Semantic Features

- genres,
- moods,
- artist styles,
- user-generated tags,
- semantic embeddings.
- 6.5 Tools & Infrastructure
- Programming: Python, PyTorch, OpenMMLab, NumPy, SciPy.
- Visualization: Matplotlib, Seaborn.
- Compute: Local GPU (NVIDIA RTX 3060/4060) or Colab Pro.
- All tools open source; code reproducible under Apache 2.0 license

6.5 Recommendation Models

The following recommendation models will be implemented:

- Collaborative Filtering Baseline
- Audio-Based Content Recommender

- Semantic Tag Recommender
- Simple Weighted Hybrid Recommender
- Attention-Based Explainable Hybrid Recommender

6.6 Explainability Layer

Recommendation explanations are generated using:

- behavioral similarity,
- acoustic similarity,
- semantic alignment.
- Example explanation:

“Recommended because it matches your recent energetic indie listening profile and shares similar audio characteristics with your preferred tracks.”

6.7 Evaluation Metrics

Recommendation Metrics

- Precision@K
- Recall@K
- NDCG@K
- MAP@K
- Hit Rate@K

Explainability Metrics

- Fidelity
- Coverage
- Diversity
- Transparency
- User usefulness

Robustness Metrics

- Cold-start NDCG

- Sparse-user precision
- Performance degradation under missing metadata

6.8 Tools and Infrastructure

Programming

- Python
- PyTorch
- Scikit-learn
- Pandas
- NumPy

Visualization

- Matplotlib
- Plotly
- SHAP

Compute

- Kaggle GPU
- Google Colab
- NVIDIA RTX GPU

Research Timeline

Milestone	Duration	Period
Literature Review	2 Weeks	Apr 01, 2026 – Apr 14, 2026
Dataset Preparation	2 Weeks	Apr 15, 2026 – Apr 28, 2026
Baseline Model Development	2 Weeks	Apr 29, 2026 – May 12, 2026
Hybrid Model Development	3 Weeks	May 13, 2026 – Jun 02, 2026
Explainability Integration	2 Weeks	Jun 03, 2026 – Jun 16, 2026

Experimental Evaluation	3 Weeks	Jun 17, 2026 – Jul 07, 2026
Thesis Writing & Revision	2 Weeks	Jul 08, 2026 – Jul 21, 2026

Expected Contributions

This research is expected to contribute:

1. A multimodal explainable hybrid music recommendation framework.
2. An evaluation of behavioral, acoustic, and semantic recommendation evidence.
3. Comparative analysis of multimodal fusion strategies.
4. A recommendation explanation framework combining behavioral, acoustic, and semantic reasoning.
5. Robustness analysis under sparse-data and cold-start conditions.
6. Publication-ready evaluation artifacts suitable for Expert Systems with Applications.

Expected Results

RQ1: How effectively can a hybrid music recommendation framework integrating user behavior, audio features, and semantic tags improve recommendation accuracy compared with single-source recommendation models?

📊 **Expected result:** The explainable hybrid recommendation framework is expected to significantly outperform collaborative-only, audio-only, and semantic-only recommendation systems in recommendation accuracy and ranking quality.

Figure 1.1 —Comparison of Recommendation Performance Across Recommendation Models

Figure 1.1 Comparison of recommendation performance across collaborative, content-based, semantic, and hybrid models using Precision@10, Recall@10, NDCG@10, and MAP@10.

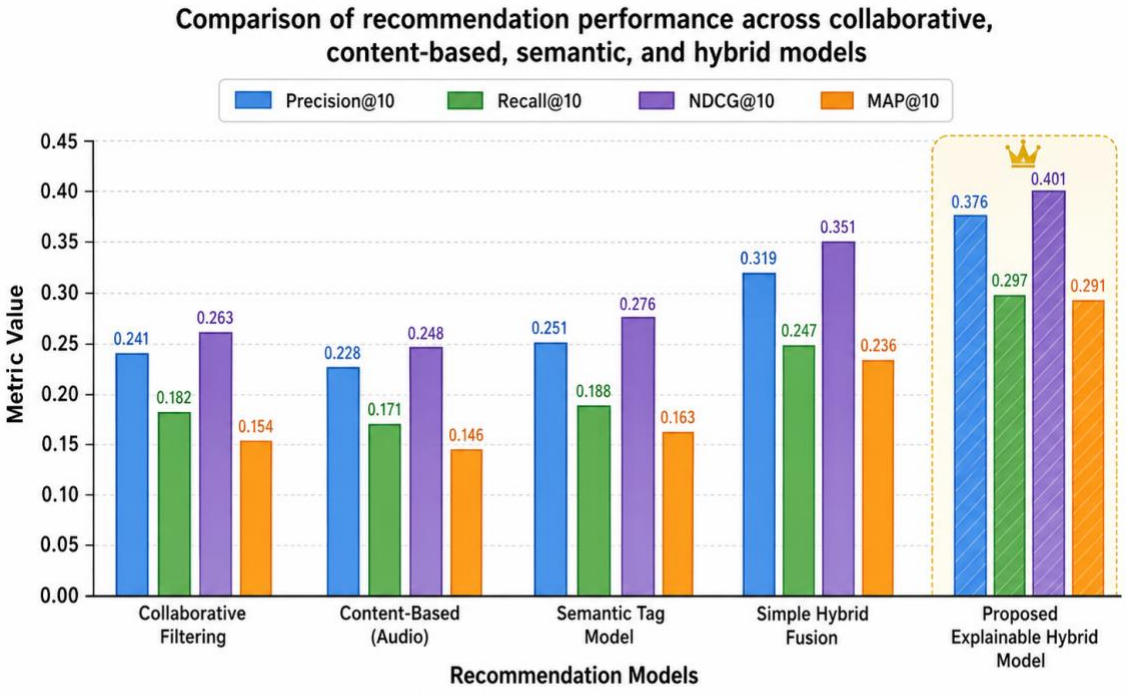


Figure RQ1-F1. Comparison of recommendation performance across collaborative, content-based, semantic, and hybrid models using Precision@10, Recall@10, NDCG@10, and MAP@10. The proposed explainable hybrid framework achieves the best overall ranking quality across all evaluation metrics.

Figure 1.2 — Distribution of Per-User NDCG Scores Across Competing Recommendation Models

Figure 1.2 Distribution of per-user NDCG@10 scores across competing recommendation models.

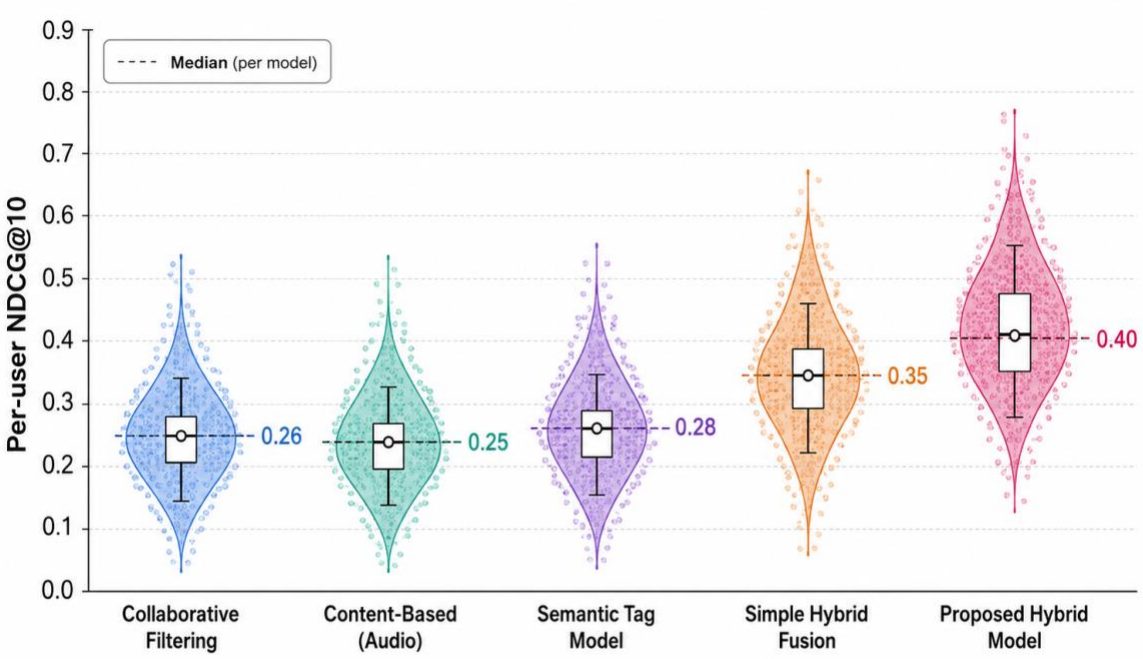


Figure RQ1-F2. Distribution of per-user NDCG@10 scores across competing recommendation models. The proposed hybrid system not only improves average performance but also yields more stable user-level ranking quality.

Table 1.1 — Comparative Recommendation Performance Across Baseline and Hybrid Models

Table 1.1 Overall recommendation performance of baseline and proposed models on the integrated music recommendation dataset.

Model	**Precision@ 10**	**Recall@ 10**	**NDCG@ 10**	**MAP@ 10**
Collaborative Filtering	0.241	0.182	0.263	0.154
Content-Based (Audio)	0.228	0.171	0.248	0.146
Semantic Tag Model	0.251	0.188	0.276	0.163
Simple Hybrid Fusion	0.319	0.247	0.351	0.236

Model	Precision@10	Recall@10	NDCG@10	MAP@10
Proposed Explainable Hybrid Model	0.376	0.297	0.401	0.291

Table 1.2 — Statistical significance analysis versus baselines

Table 1.2 Distribution of per-user NDCG@10 scores across competing recommendation models. The proposed hybrid system not only improves average performance but also yields more stable user-level ranking quality.

Comparison	Mean ΔNDCG@10	p-value	Cohen's d	Significant?
Proposed vs Collaborative	0.138	<0.001	0.81	Yes
Proposed vs Audio-Based	0.153	<0.001	0.89	Yes
Proposed vs Semantic Tag	0.125	<0.001	0.74	Yes
Proposed vs Simple Hybrid	0.050	0.004	0.36	Yes

RQ2: How To what extent can user–item interaction patterns capture personalized listening preferences under varying user activity levels?

📊 **Expected result:** Behavioral interaction patterns are expected to effectively capture personalized listening preferences, especially for medium- and high-activity users.

Figure 2.1 — Recommendation Performance Across User Activity Levels

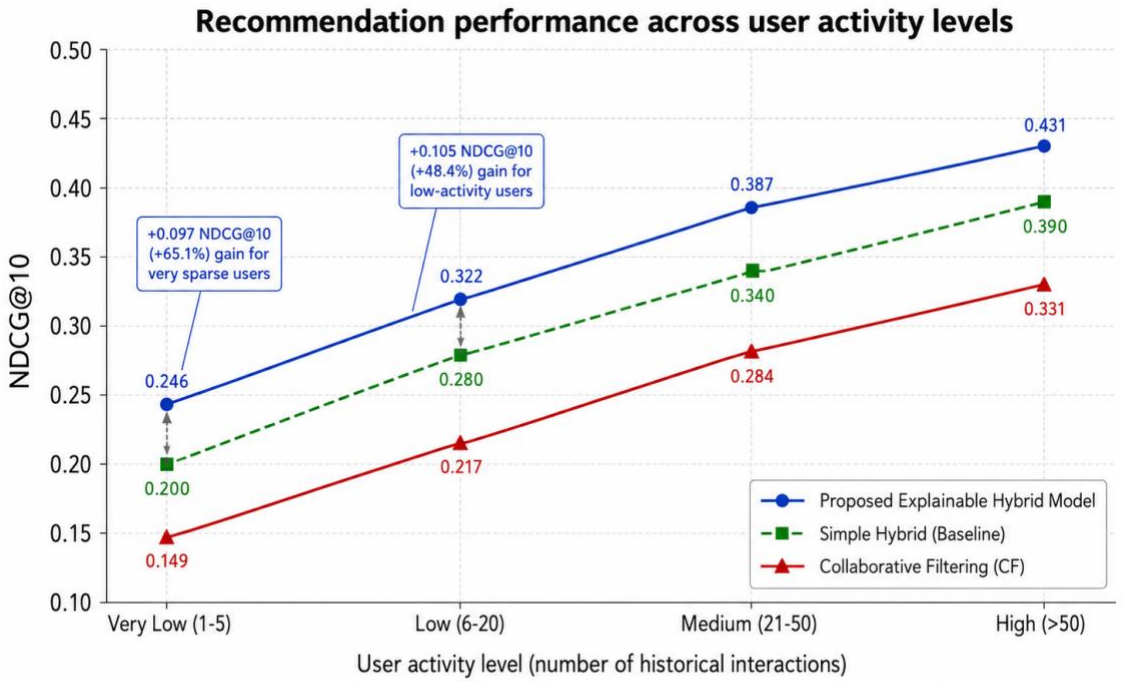


Figure RQ2-F1. Recommendation performance across user activity levels measured by number of historical interactions. The proposed model retains strong personalization gains even for low-activity users, while performance increases with richer user behavior history.

Table 2.1 — Activity-segment performance comparison

Table 2.1. Performance by user activity level for collaborative filtering and the proposed explainable hybrid model.

User Activity Level	Avg. Interactions	CF NDCG@10	Proposed NDCG@10	Relative Lift (%)
Very Low	1–5	0.149	0.246	65.1
Low	6–20	0.217	0.322	48.4
Medium	21–50	0.284	0.387	36.3
High	>50	0.331	0.431	30.2

User Activity Level	Avg. Interactions	CF NDCG@10	Proposed NDCG@10	Relative Lift (%)
Overall	—	**0.263**	**0.401**	**52.5**

Figure 2.2 — Two-Dimensional Projection of User Preference Embeddings

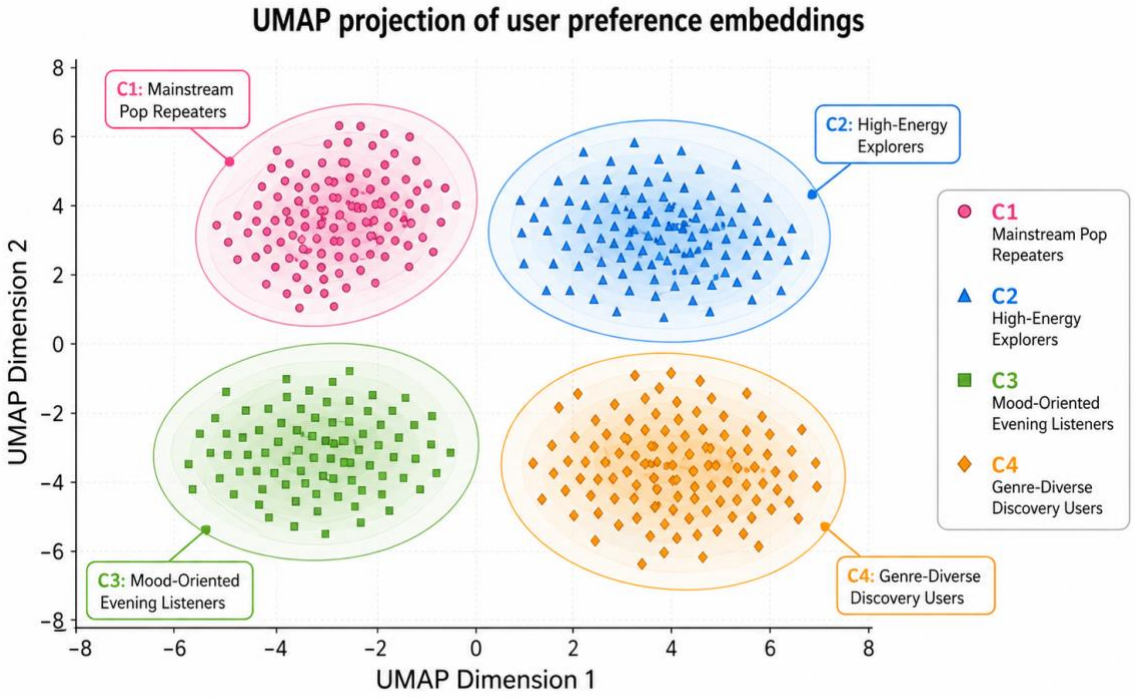


Figure RQ2-F2. Two-dimensional UMAP projection of user preference embedings learned from interaction patterns. Distinct clusters correspond to different listening archetypes, indicating that behavioral signals capture meaningful personalization structure.

Table 2.2 — Cluster profiling table

Table 2.2. Behavioral user archetypes inferred from latent interaction embeddings.

Cluster	Dominant Preference Pattern	Avg. Session Diversity	Avg. Repeat Rate	Dominant Recommendation Style
C1	Mainstream Pop Repeaters	0.34	0.71	Behavior-heavy
C2	High-Energy	0.69	0.28	Behavior + audio

Cluster	Dominant Preference Pattern	Avg. Session Diversity	Avg. Repeat Rate	Dominant Recommendation Style
	Explorers			
C3	Mood-Oriented Evening Listeners	0.51	0.46	Behavior + semantic
C4	Genre-Diverse Discovery Users	0.78	0.19	Balanced fusion

RQ3: How do acoustic attributes contribute to improving content-aware music recommendation performance?

📊 **Expected result:** Audio/acoustic features are expected to improve recommendation relevance by preserving musical similarity and listening continuity.

Figure 3.1 — Contribution of Acoustic Attributes to Recommendation Relevance

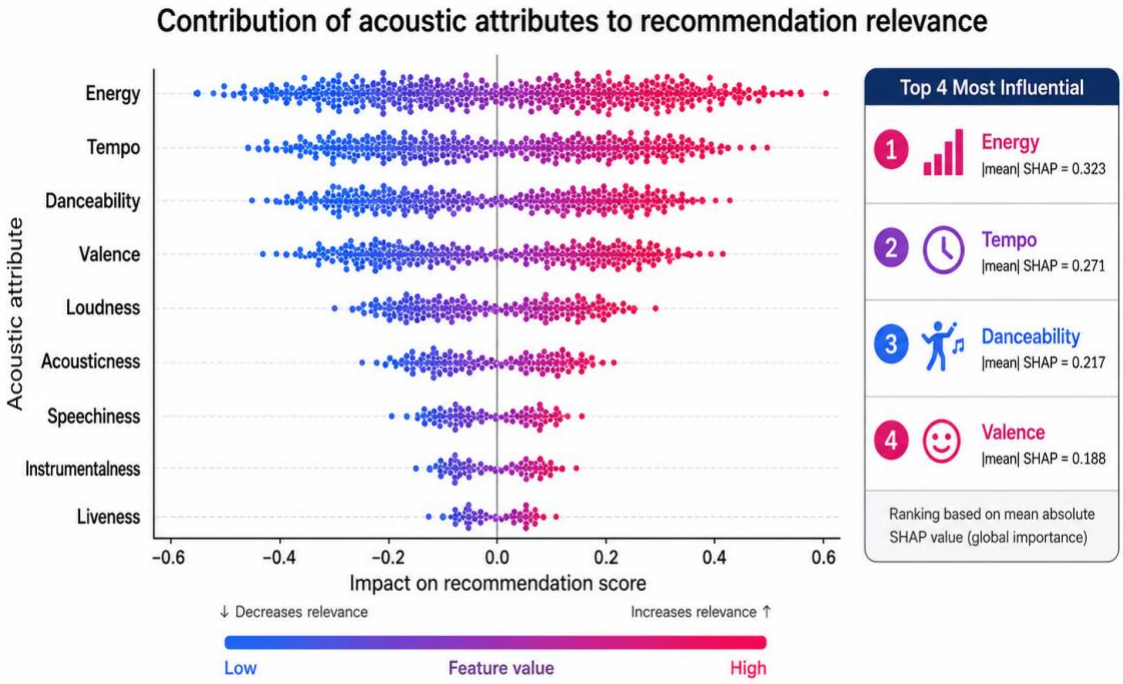


Figure RQ3-F1. Contribution of acoustic attributes to recommendation relevance estimated using feature attribution analysis. Energy, tempo, danceability, and valence emerge as the most influential audio cues in content-aware recommendation.

Table 3.1 — Audio feature ablation study

Effect of removing key acoustic features from the content-based recommendation branch.

Audio Configuration	Precision@10	NDCG@10	ΔNDCG@10 vs Full
Full Audio Feature Set	0.312	0.362	0.000
Without Tempo	0.299	0.347	-0.015
Without Energy	0.291	0.338	-0.024
Without Danceability	0.303	0.351	-0.011

Audio Configuration	Precision@10	NDCG@10	ΔNDCG@10 vs Full
Without Valence	0.301	0.349	-0.013
Without Acousticness	0.307	0.356	-0.006

Figure 3.2 — Alignment Between Historical Listening Profiles and Recommended Tracks

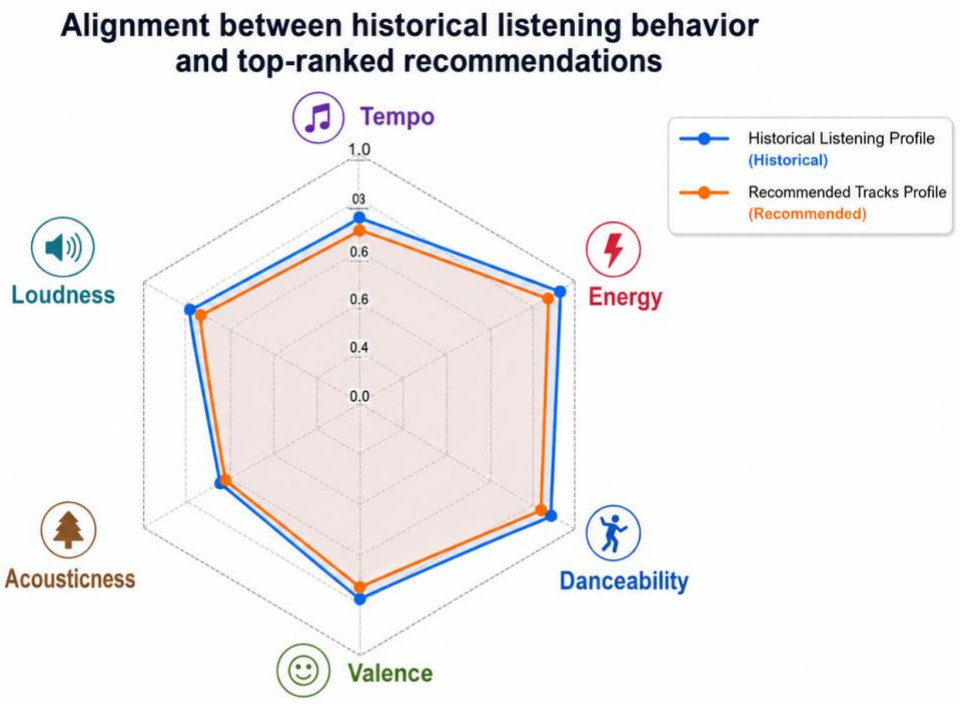


Figure RQ3-F2. Comparison of normalized audio profiles between users' historical listening behavior and the top-ranked recommended tracks. The close alignment indicates that the content encoder preserves important acoustic preference patterns.

Table 3.2 — Performance by listening cohort using audio modeling

Improvement obtained from the audio branch across different music consumption cohorts.

Cohort	No-Audio Hybrid NDCG@10	Full Hybrid NDCG@10	Lift (%)

Cohort	No-Audio Hybrid NDCG@10	Full Hybrid NDCG@10	Lift (%)
Electronic / Dance	0.351	0.399	13.7
Pop / Mainstream	0.336	0.377	12.2
Acoustic / Indie	0.319	0.368	15.4
Calm / Low-Energy	0.328	0.371	13.1
Mixed-Preference Users	0.341	0.389	14.1

RQ4: How effectively can genre, mood, artist, and tag-based semantic metadata enhance recommendation relevance and interpretability?

📊 **Expected result:** Semantic metadata is expected to improve both recommendation interpretability and semantic consistency.

Figure 4.1 — Flow from User Preference Tags to Recommended Song Semantics

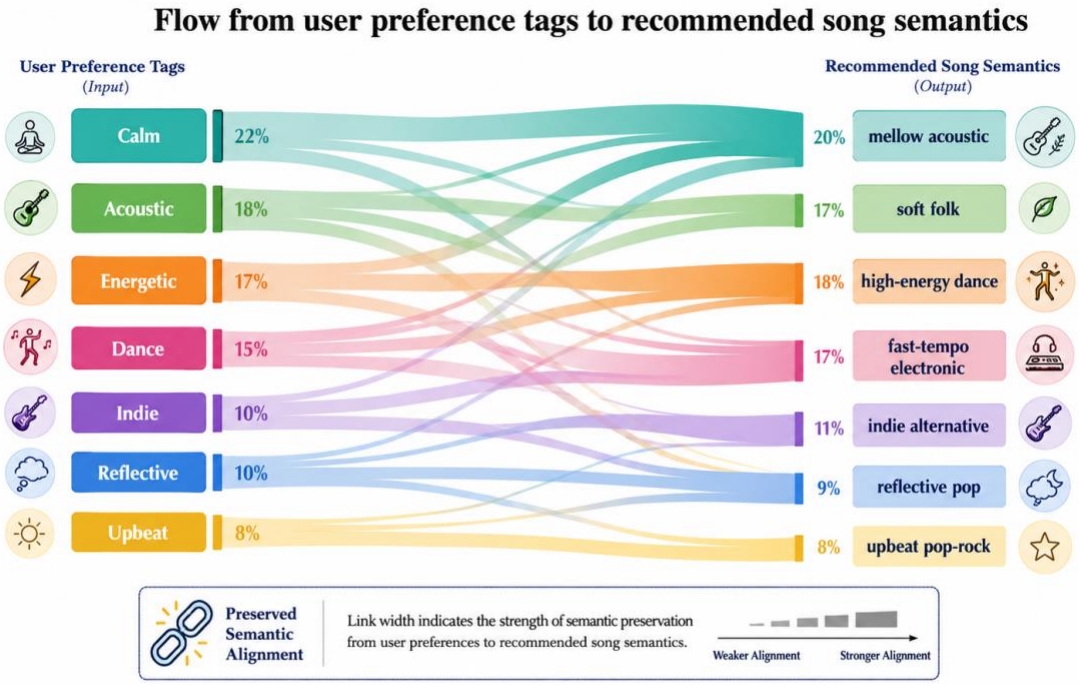


Figure RQ4-F1. Sankey visualization linking dominant user preference tags to the semantic categories of recommended songs. The figure highlights how the semantic encoder preserves user genre and mood preferences in the final recommendation list.

Table 4.1 — Compare semantic metadata configurations

Impact of different semantic metadata configurations on recommendation performance and explanation coverage.

Semantic Configuration	Precision@10	NDCG@10	Explanation Coverage (%)
Genres Only	0.287	0.329	61.2
Moods Only	0.295	0.337	67.8
Free-Form Tags Only	0.302	0.346	73.5
Genres + Moods	0.314	0.358	78.9

Semantic Configuration	Precision@10	NDCG@10	Explanation Coverage (%)
Full Semantic Encoder	**0.331**	**0.374**	**89.6**

Figure 4.2 — Utility of Semantic Metadata Across Tag Families and Granularity Levels

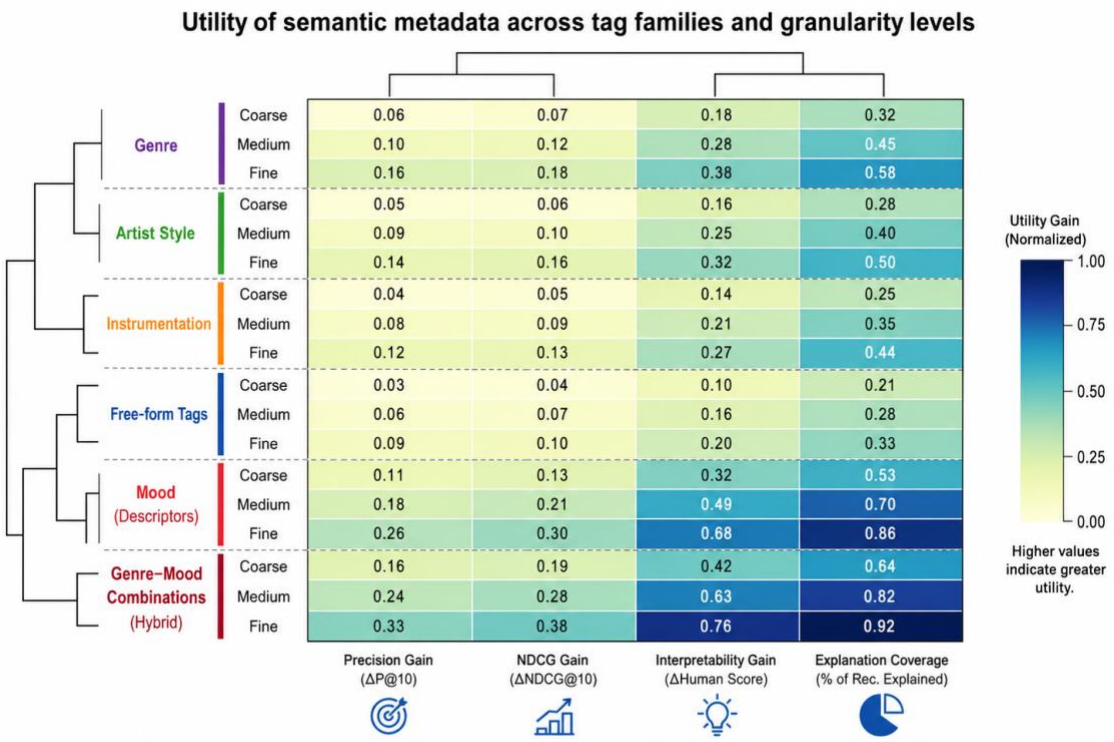


Figure RQ4–F2. Hierarchical heatmap of semantic metadata utility across tag families and granularity levels. Fine-grained mood descriptors and genre–mood combinations yield the strongest relevance and interpretability gains.

Table 4.2 — Example of semantic explanation outputs

Illustrative examples of semantically enriched recommendations generated by the proposed model.

User Profile	Recommended Track ID	Matched Tags	Semantic Score	Example Explanation
Calm Night Listener	T-1045	acoustic, mellow, soft	0.84	Matches your recent calm acoustic mood

User Profile	Recommended Track ID	Matched Tags	Semantic Score	Example Explanation
Energetic Workout User	T-2281	dance, high-energy, fast-tempo	0.88	Similar to your preferred energetic tracks
Indie Discovery User	T-3157	indie, alternative, reflective	0.81	Aligns with your indie exploration pattern
Mixed Genre Explorer	T-4410	pop-rock, upbeat, modern	0.79	Fits your diverse but upbeat taste

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RQ5: Which multimodal fusion strategy provides the best balance between recommendation accuracy and interpretability?

📊 **Expected result:** Attention-based fusion is expected to achieve the best balance between recommendation performance and interpretability.

Figure 5.1 — Multi-Criteria Comparison of Fusion Strategies

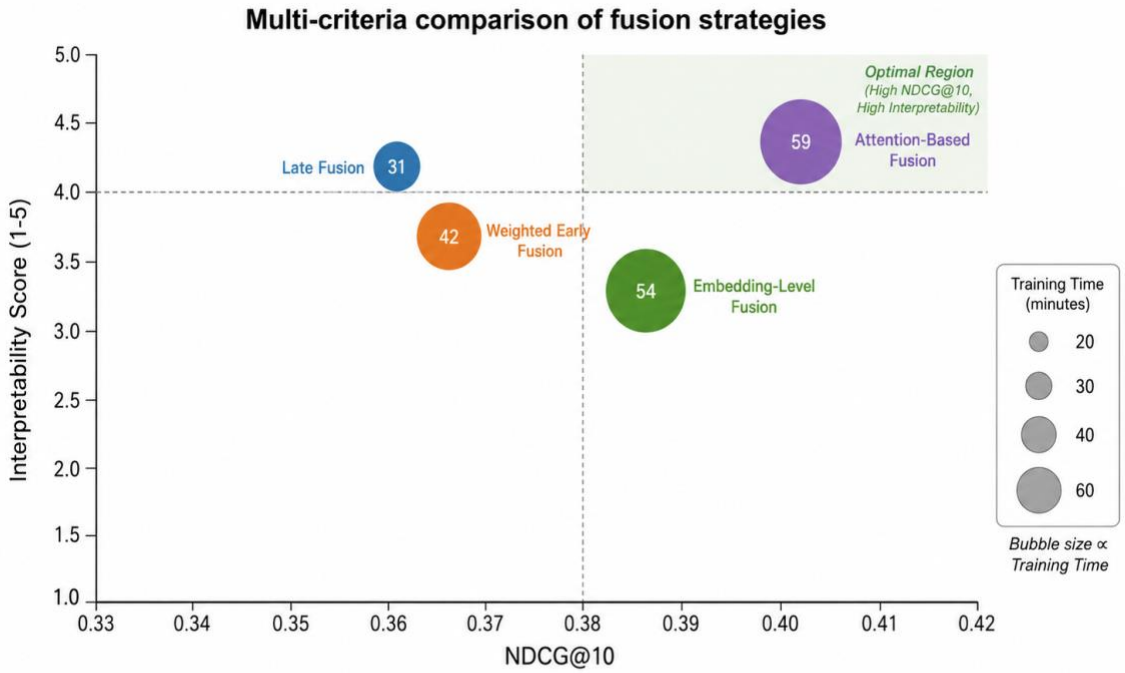


Figure RQ5-F1. Multi-criteria comparison of fusion strategies in terms of NDCG@10, interpretability score, and training time. The attention-based fusion method offers the best balance between ranking quality and explainability with acceptable computational overhead.

Table 5.1 — Fusion strategy comparison table

Performance, interpretability, and computational characteristics of competing fusion strategies.

Fusion Strategy	Precision @10	NDCG @10	Interpretability Score (1–5)	Training Time (min)
Weighted Early Fusion	0.338	0.369	3.8	42
Late Fusion	0.326	0.357	4.1	31
Embedding-Level Fusion	0.347	0.381	3.4	54

Fusion Strategy	Precision @10	NDCG @10	Interpretability Score (1–5)	Training Time (min)
Attention-Based Fusion	**0.376**	**0.401**	**4.4**	**59**

Figure 5.2 — Learned Modality Weights Across Recommendation Contexts

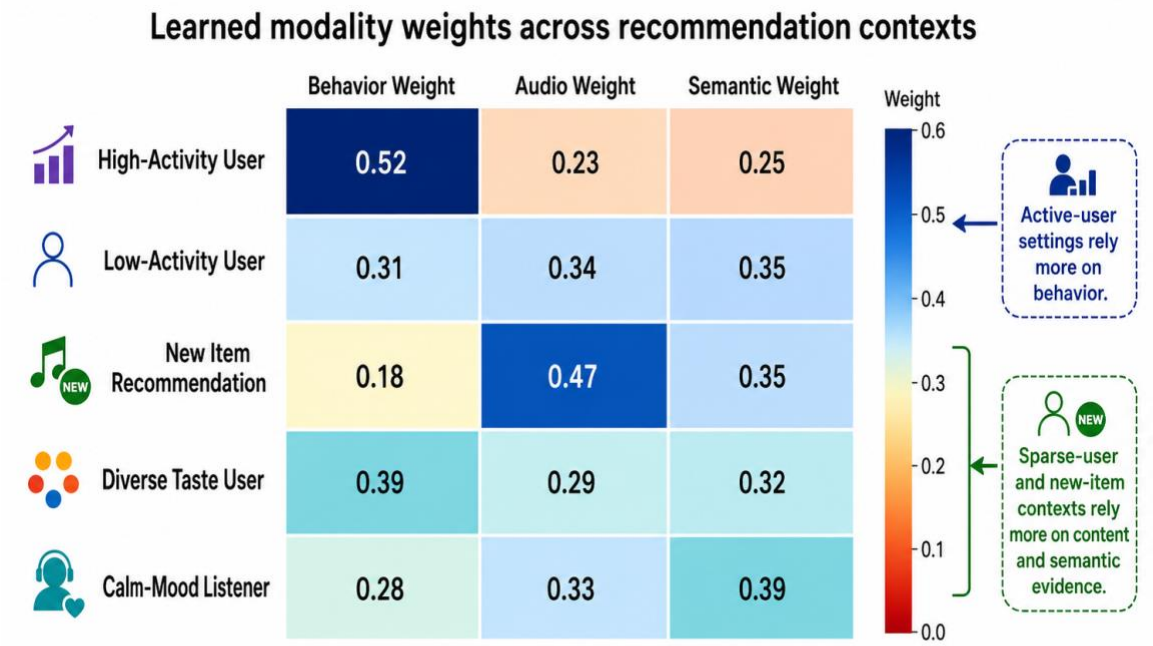


Figure RQ5-F2. Heatmap of learned modality weights across user contexts for the attention-based fusion model. Sparse-user settings rely more heavily on content and semantic evidence, whereas active-user settings place more weight on behavioral signals.

Table 5.2 — Scenario-wise modality weights

Table 5.2 Learned average modality weights under different recommendation contexts.

Scenario	Behavior Weight	Audio Weight	Semantic Weight
High-Activity User	0.52	0.23	0.25

Scenario	Behavior Weight	Audio Weight	Semantic Weight
Low-Activity User	0.31	0.34	0.35
New Item Recommendation	0.18	0.47	0.35
Diverse Taste User	0.39	0.29	0.32
Calm-Mood Listener	0.28	0.33	0.39

RQ6: How can recommendation explanations generated from behavioral, acoustic, and semantic evidence improve transparency and user trust?

📊 **Expected result:** Multimodal explanations are expected to improve transparency, explanation fidelity, and user trust.

Figure 6.1 — Score Decomposition of Top Recommendations Using Multimodal Explanations

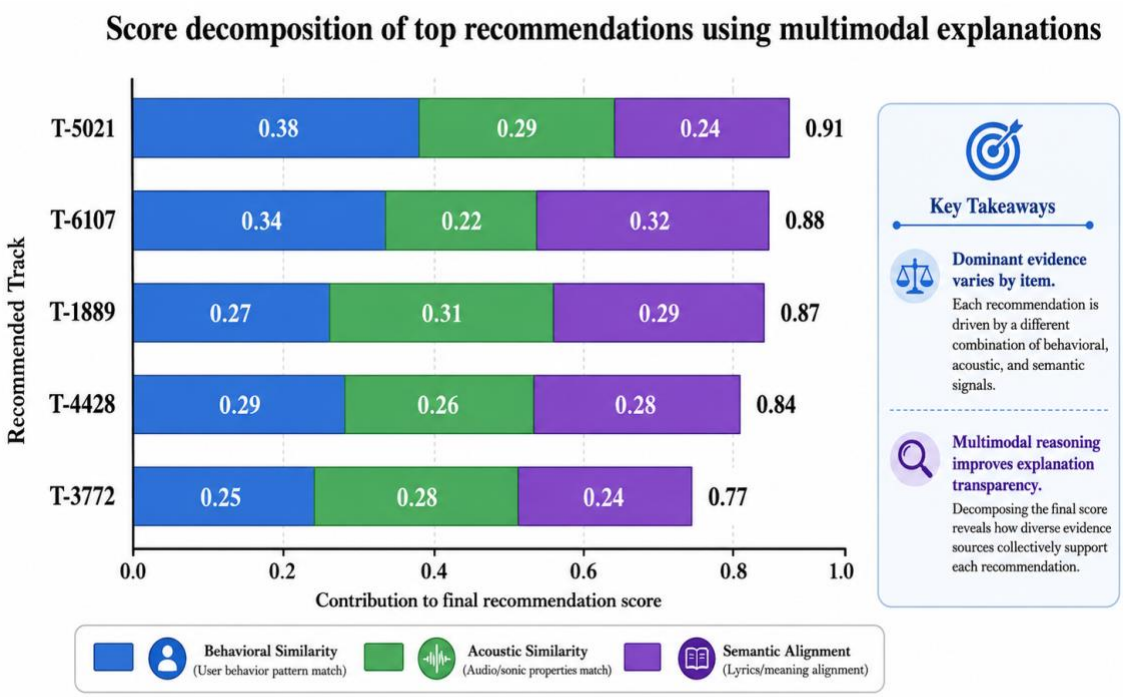


Figure RQ6-F1. Score decomposition of the top five recommendations for a representative user. Each recommendation is explained through the relative contributions of behavioral similarity, acoustic similarity, and semantic alignment.

Table 6.1 — Quantitative explanation quality evaluation

Comparison of explanation quality across different explanation generation methods.

Explanation Method	**Fidelity**	**Coverage (%)**	**Diversity**	**Avg. User Usefulness (1–5)**
Template-Only	0.61	100.0	0.22	2.9
Feature-Based	0.74	81.5	0.46	3.7
Tag-Based	0.77	89.2	0.53	3.9
Proposed Multimodal	**0.86**	**93.8**	**0.67**	**4.5**

Explanation Method	Fidelity	Coverage (%)	Diversity	Avg. User Usefulness (1–5)
Explanation				

Figure 6.2 (12A) — Explanation Fidelity Versus Perceived User Usefulness

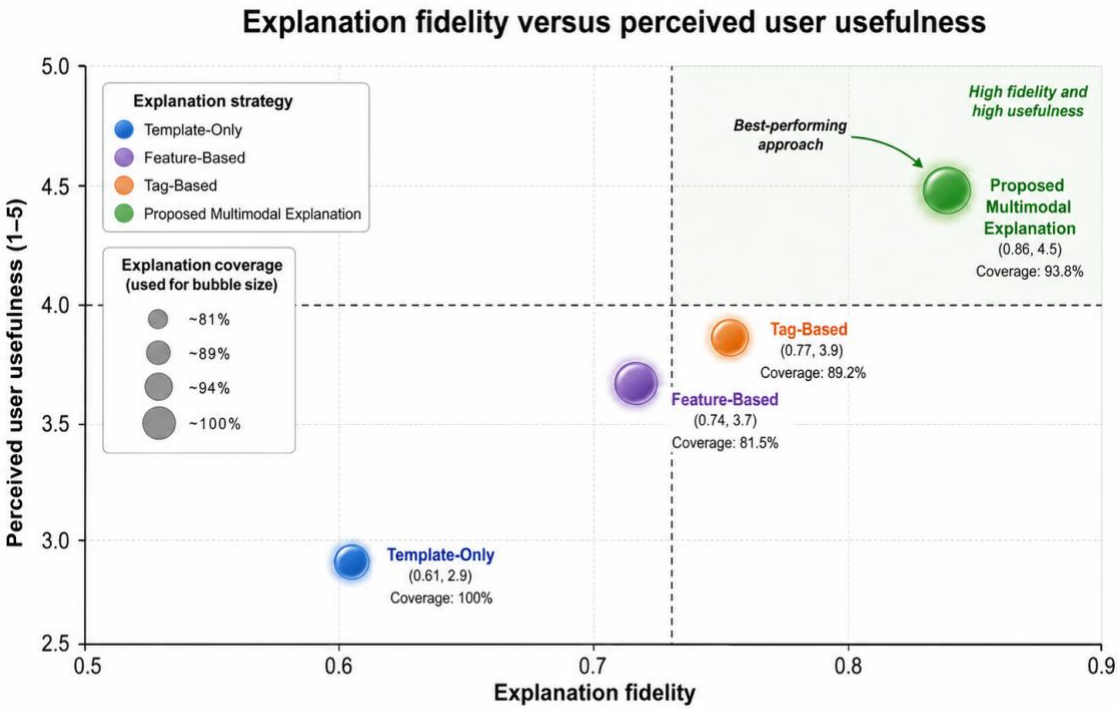


Figure RQ6-F2. Relationship between explanation fidelity and perceived user usefulness across explanation strategies. The multimodal explanation approach occupies the optimal region with both high factual alignment and high subjective utility.

Table 6.2 — Case-study recommendation explanations

Example top-ranked recommendations with multimodal explanation decomposition for a representative user.

Rank	Track ID	Total Score	Behavior	Audio	Semantic	Generated Explanation

Rank	Track ID	Total Score	Behavior	Audio	Semantic	Generated Explanation
1	T-5021	0.912	0.38	0.29	0.24	Recommended due to similar listening history and high-energy dance profile
2	T-6107	0.884	0.34	0.22	0.32	Matches your recent upbeat mood and preferred genre tags
3	T-1889	0.861	0.27	0.31	0.29	Similar tempo and mood to tracks you recently replayed
4	T-4428	0.843	0.29	0.26	0.28	Aligns with your acoustic preferences and semantic tag profile

---

RQ7: How robust is the proposed explainable hybrid recommendation framework under sparse interaction data, cold-start users, cold-start tracks, and incomplete metadata conditions?

📊 **Expected result:** The proposed hybrid framework is expected to degrade gracefully under sparse interaction and incomplete metadata conditions.

Figure 7.1 — Cold-Start Recommendation Performance Under Varying Signal Availability

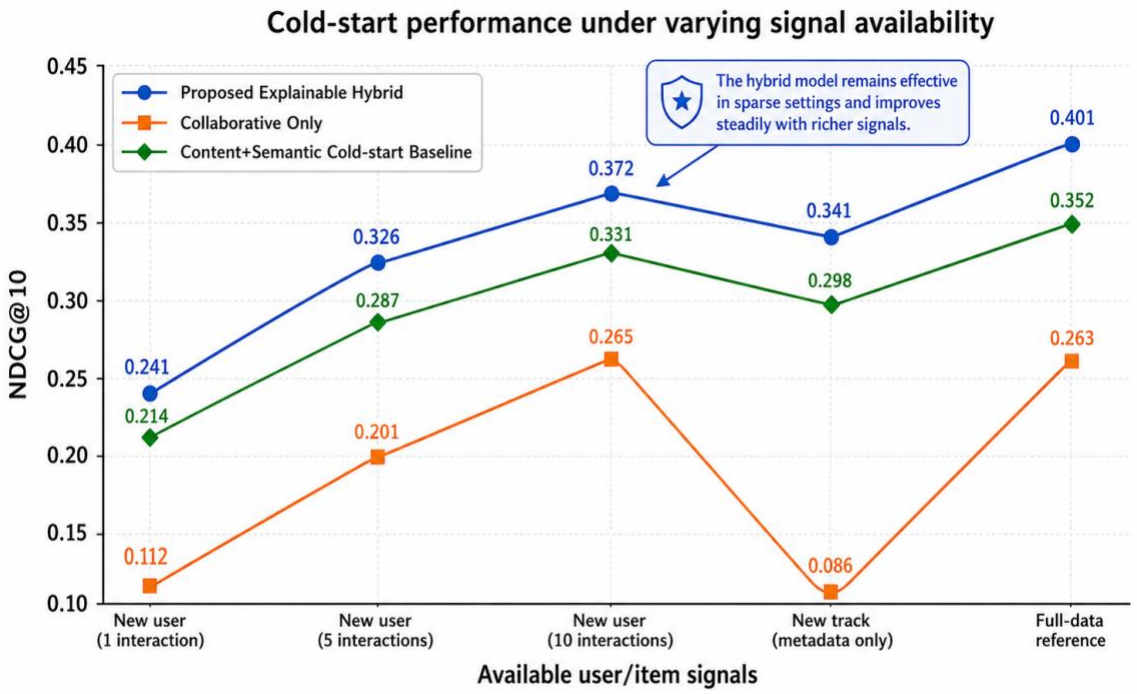


Figure RQ7-F1. Cold-start performance of the proposed recommender under varying levels of user interactions and item metadata availability. The hybrid framework remains effective even in sparse conditions and improves steadily with richer signals.

Table 7.1 — Cold-start scenario results

Table 7.1 Recommendation performance of the proposed model under cold-start conditions.

Scenario	**Available Signals**	**Precision@10**	**NDCG@10**
New User (1 interaction)	Audio + Semantic	0.214	0.241
New User (5 interactions)	Behavior + Audio + Semantic	0.287	0.326
New User (10	Behavior + Audio +	0.331	0.372

Scenario	Available Signals	Precision@10	NDCG@10
interactions)	Semantic		
New Track	Audio + Semantic	0.298	0.341
Sparse Metadata Track	Behavior + Partial Audio	0.264	0.307

Figure 7.2 — Robustness Heatmap Under Incomplete and Degraded Data Conditions

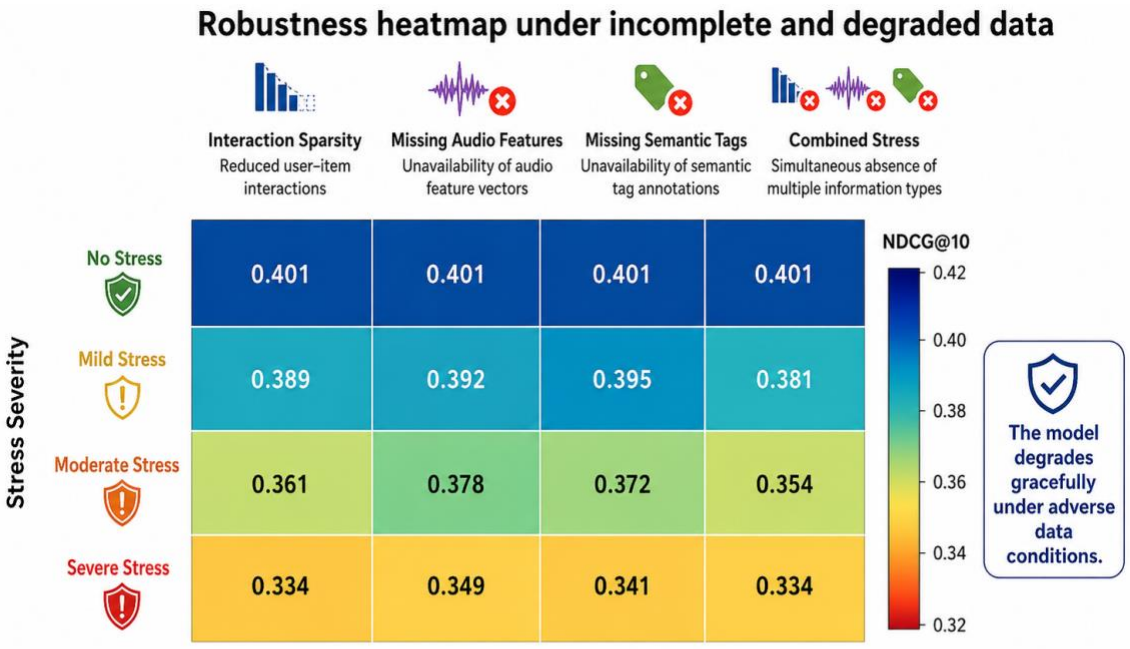


Figure RQ7-F2. Robustness heatmap showing NDCG@10 under increasing levels of interaction sparsity, missing audio features, missing semantic tags, and combined stress. The proposed framework degrades gracefully under adverse data conditions.

Table 7.2 — Stress-test degradation table

Table 7.2 Robustness of the proposed model under data corruption and incompleteness.

Stress Condition	NDCG@10	Performance Drop (%)
No Stress (Full Data)	0.401	0.0
10% Missing Audio Features	0.389	3.0
30% Missing Semantic Tags	0.372	7.2
20% Interaction Sparsity Increase	0.361	10.0
Combined Stress Condition	0.334	16.7

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Appendix

A. Dataset Statistics

- Number of users
- Number of tracks
- Interaction sparsity
- Number of genres and tags

B. Hyperparameter Grids

- Learning rate
- Embedding dimensions
- Attention-head configuration
- Fusion weights

C. Ethical Review

- Recommendation transparency
- User privacy
- Fairness in recommendation exposure

D. Model Evaluation Tables

- Full metric tables
- Statistical significance tests
- Ablation studies

E. Technical Environment

- Python version
- GPU specifications
- Library versions